Displacement

If I cannot rebuild it, I do not yet understand it.

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What is a displacement?
The change in position from the initial coordinate to the final coordinate, represented as a straight arrow starting from the initial coordinate and finishing at the final one (see Figure 1).

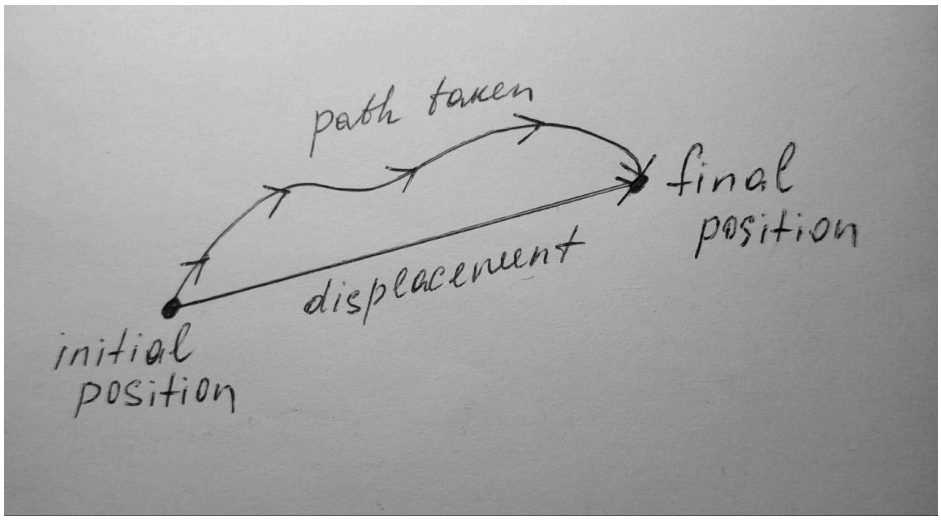

Figure 1

<u>Note</u> that I was thinking before that the displacement is the shortest path CONSIDERING the obstacles. Imagine the initial and final points, with an obstacle (a rectangle) between them. I thought displacement would avoid the obstacle, moving around it, but in reality it passes straight through it (see Figure 2).

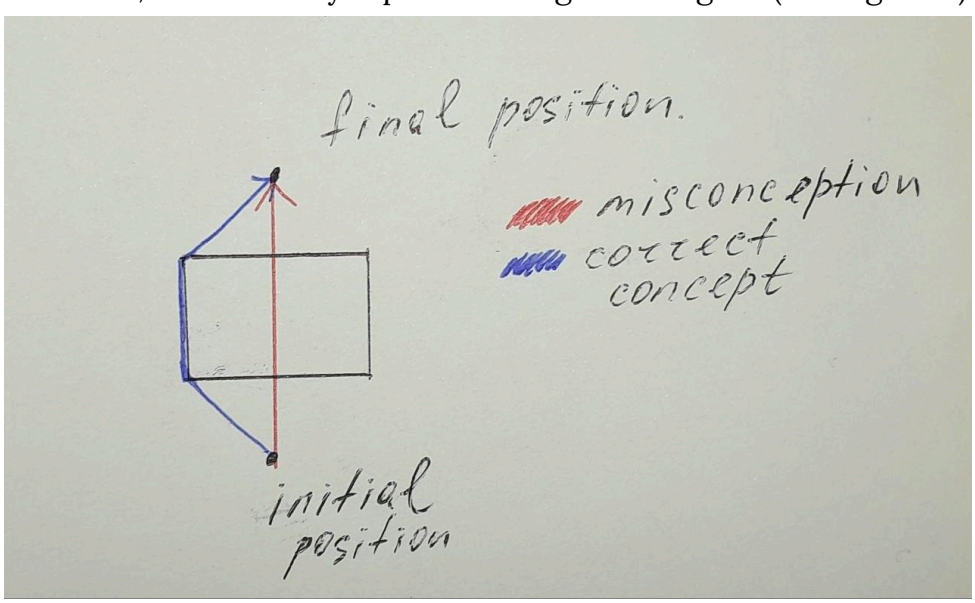

Figure 2

What is the total path? It is the length of the path travelled.
<u>*How would I define them?*</u>
<u>*(formulas)*</u>

for the displacement: $\Delta x = x_{final} - x_{initial}$ or $\int dx$ $(or \int vdt)$

for the total path: $\sum_{i=1}^{n} |x_i - x_{i-1}|$ or $\int |dx|$ $(or \int |vdt|)$

and *it is possible that distance travelled does not equal displacement.*

<u>*These are the questions that came to my mind:*</u>
1. *Can displacement be negative?*
Yes. Imagine a ball, and x - axis. The ball first goes a little to the right, then a lot to the left. That is where displacement is negative (see Figure 3).
and it is 0 when we return back to our initial position

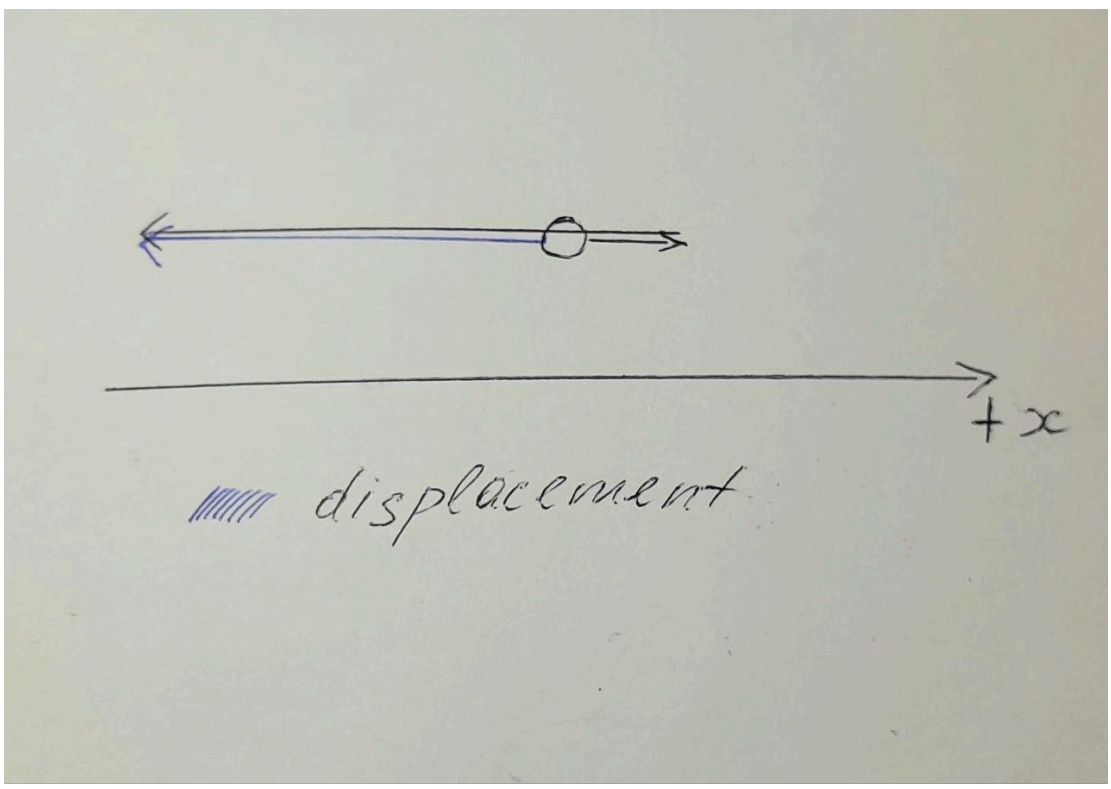

Figure 3

2. *overall, why* S = *v*t? Why not v/t?*

When the velocity is constant, we say it is the distance travelled per unit of time. So intuitively, if v = 5 meters per second:

The first second -> 5 meters

The first 2 seconds total path -> 10 meters.

Why not divide?

S1 = 5/1 = 5 (seems okay)

S2 = 5/2 = 2.5

S3 = 5/5 = 1 meters – this assumes that as time increases, the distance travelled decreases. Which is wrong.

And the same thing is for + and - signs (so S ≠ v + t or S ≠ v - t).

3. *And ∫v * dt - why not v(t)?* because the velocity is constant. So we don't need even integrate, and we directly write v*t (since v is constant, it can be taken out of the integral, and the integral of dt over the interval gives the total time t). But if velocity changed, I would use v(t) for finding the specific velocity for that time. Because at the smallest amounts, even if the overall velocity changes, the smallest amounts are constant. This is like drawing a circle: it looks circular overall. But this circle is made of very little straight arrows (see Figure 4).

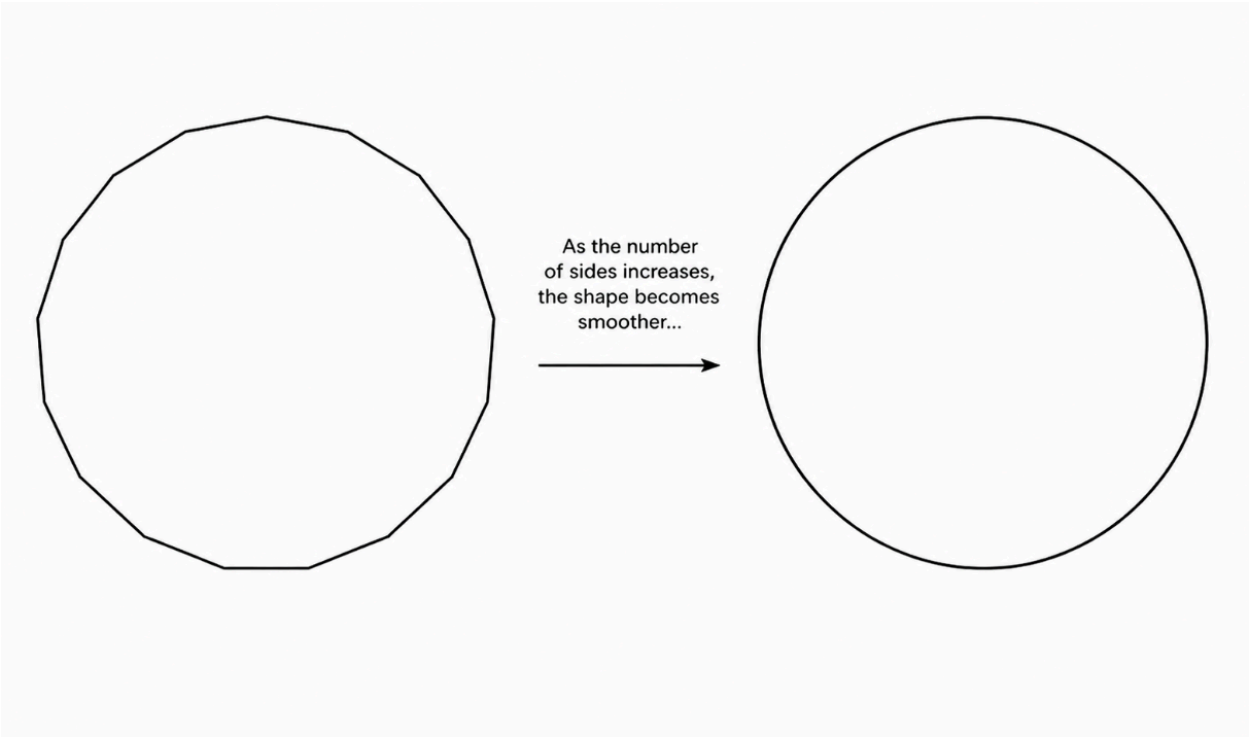
As the number
of sides increases,
the shape becomes
smoother...

Figure 4